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QUANTIFYING MECHANISMS OF BACTERIAL PHOSPHATE ECONOMY FOR SYNTHETIC APPS

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Escherichia coli have developed one of the most efficient regulatory response mechanisms to phosphate starvation. The machinery involves a cascade with a two-component system (TCS) that relays the external signal to the genetic circuit, resulting in a feedback response. Achieving a quantitative understanding of this system has implications in synthetic biology and biotechnology, for example, in applications for wastewater treatment. We present a computational model and experimental results with a detailed description of the TCS, consisting of PhoR and PhoB, together with the mechanisms of gene expression [1]. The model is parameterised within the feasible range, and fitted to the dynamic response of our experimental data on PhoB as well as PhoA, the product of this network that is used in alkaline phosphatase production. Deterministic and stochastic simulations with our model predict the regulation dynamics in higher external phosphate concentrations while reproducing the experimental observations. Flux analysis on stochastic simulations reveal the distribution of the system resources in response to starvation signal [2]. In a cycle of simulations and experimental verification, our model predicts and explores phenotypes with various synthetic promoter designs that optimise the inorganic phosphate intake in *E. coli*. Our results suggest directions for investigations on engineering new sensors and regulators for living technologies.

References

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